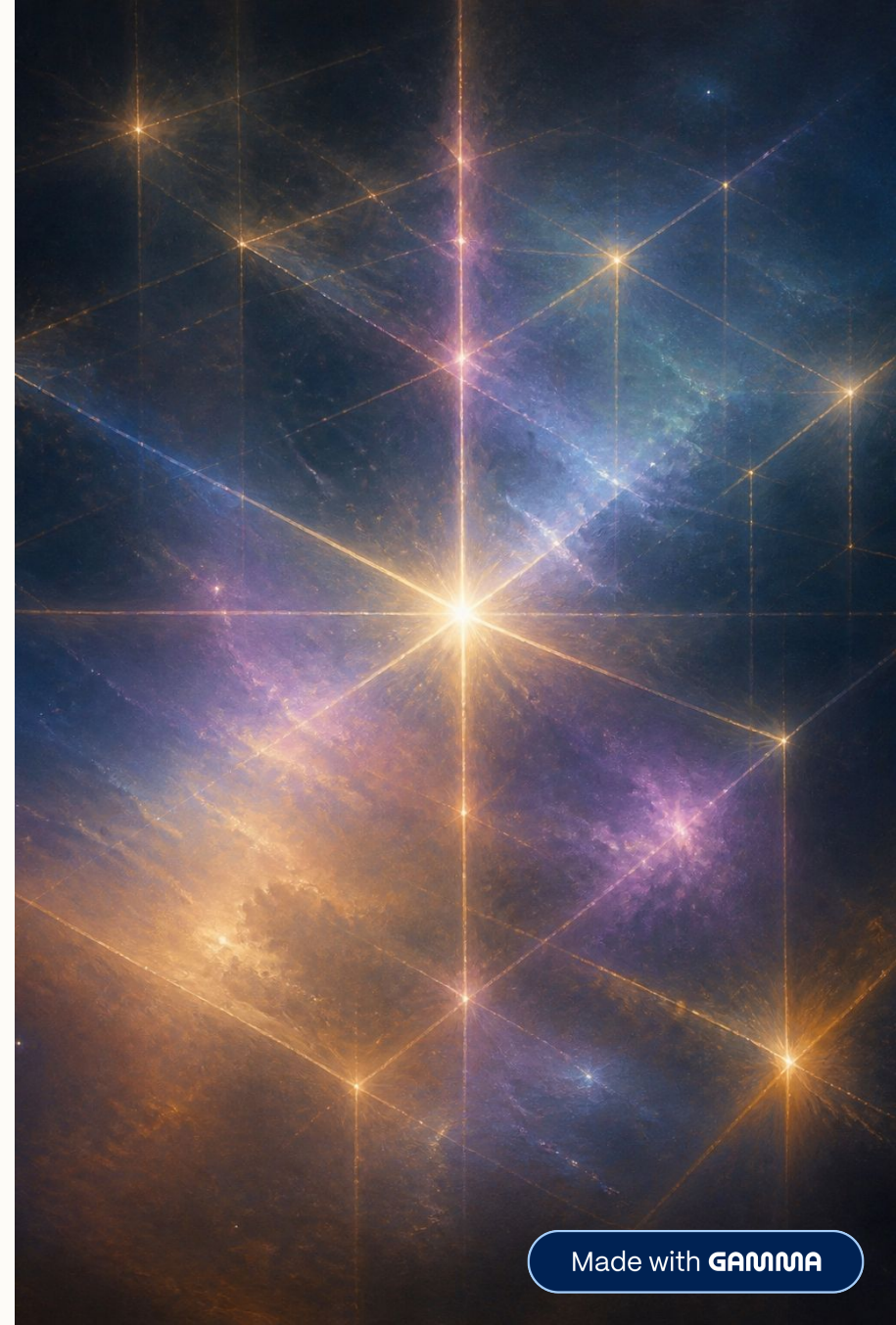


Hilbert Spaces & Inner Products

The mathematical language of quantum mechanics — where vectors, geometry, and probability converge to describe the quantum world.



What Is a Hilbert Space?



A **Hilbert Space** is a **complete vector space equipped with an inner product**. It generalizes Euclidean geometry to higher — even infinite — dimensions.

Quantum States

Every quantum state is a vector in a Hilbert space.

Mathematical Backbone

Enables probability calculations and quantum measurements.

Qubit Representation

$\mathbb{C}^2$ is the Hilbert space of a single qubit.

Four Defining Properties

These axioms make Hilbert spaces the right setting for quantum mechanics.



Vector Addition

The sum of any two vectors stays in the space.



Scalar Multiplication

Multiplying by a scalar yields another valid vector.



Inner Product

Measures angles, lengths, and orthogonality between vectors.



Completeness

Every Cauchy sequence converges to a limit within the space.



Key examples: $\mathbb{R}^2$, $\mathbb{R}^3$, and $\mathbb{C}^2$ — the two-dimensional complex space representing a single qubit.

Representing Quantum States

A quantum state is encoded as a **state vector** $|\psi\rangle$. The computational basis vectors are:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Any qubit state is a **superposition**:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

The constraint ensures total probability equals 1.



The Inner Product

The inner product $\langle \phi | \psi \rangle$ maps two vectors to a complex number, encoding similarity, angle, and length.



Notation

$\langle \phi | \psi \rangle$ — the "bra-ket" or Dirac notation.

Self-overlap

$\langle 0 | 0 \rangle = 1$ — a state is perfectly aligned with itself.

Orthogonality

$\langle 0 | 1 \rangle = 0$ — basis states are mutually orthogonal.

Three Axioms of the Inner Product

1 Positivity

$\langle x|x \rangle \geq 0$, with equality only when $x = 0$. Guarantees a meaningful notion of "length."

2 Conjugate Symmetry

$\langle x|y \rangle = \langle y|x \rangle^*$. Swapping vectors takes the complex conjugate.

3 Linearity

$\langle x|ay + bz \rangle = a\langle x|y \rangle + b\langle x|z \rangle$. Linear in the second argument, antilinear in the first.

Orthogonality

Two vectors are **orthogonal** when their inner product vanishes:

$$\langle x|y\rangle = 0$$

The basis states satisfy $|0\rangle \perp |1\rangle$, meaning they are perfectly distinguishable by measurement.



State Distinguishability

Orthogonal states can be told apart with certainty.

Measurement Basis

Orthogonal bases define valid quantum measurement outcomes.

Algorithm Design

Quantum algorithms exploit orthogonality for interference and amplification.

Norm & Normalization



The **norm** (length) of a vector is defined via the inner product:

$$\|x\| = \sqrt{\langle x|x \rangle}$$

A **normalized** quantum state satisfies:

$$\langle \psi | \psi \rangle = 1$$

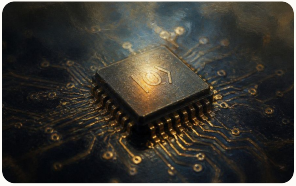
This is not merely a convention — it ensures the **total probability of all outcomes sums to 1**, a fundamental requirement of quantum mechanics.

□ Any non-zero vector can be normalized by dividing by its norm:

$$|\psi_{\text{norm}}\rangle = |\psi\rangle / \|\psi\|$$

Hilbert Spaces in Quantum Computing

Hilbert spaces underpin every layer of quantum information science.



Qubits & Gates

Qubits live in $\mathbb{C}^2$; gates are unitary operators on Hilbert space.



Algorithms & Teleportation

Shor's and Grover's algorithms rely on interference in Hilbert space.



Cryptography & Error Correction

BB84 and error-correcting codes exploit orthogonality and entanglement.



Quantum Machine Learning

Variational circuits optimize states in high-dimensional Hilbert spaces.

SUMMARY

Key Takeaways

01

Complete Inner Product Space

A Hilbert space is a complete vector space with an inner product — the stage for quantum mechanics.

02

States Are Vectors

Quantum states are unit vectors $|\psi\rangle$ satisfying $\langle\psi|\psi\rangle = 1$.

03

Inner Product = Probability

$|\langle\phi|\psi\rangle|^2$ gives the probability of measuring $|\phi\rangle$ in state $|\psi\rangle$.

04

Orthogonality = Distinguishability

Orthogonal states are perfectly distinguishable — the basis of quantum measurement.

05

Foundation of Quantum Computing

From qubits to error correction, Hilbert spaces are indispensable to quantum information.